

# Affine Coordinate-Pivot Sweeps for Root Anti-Concentration

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## 1 Theoretical Setup

### 1.1 Problem Setting

Fix an integer  $N \geq 1$ , a support radius  $R > 0$ , a density cap  $\kappa \in (0, \infty)$ , and a compact interval  $\Theta \subset \mathbb{R}$ . Let

$$b : \Theta \rightarrow \mathbb{R}, \quad F = (F_1, \dots, F_N) : \Theta \rightarrow \mathbb{R}^N, \quad \phi_\alpha(\theta) = b(\theta) + \langle \alpha, F(\theta) \rangle.$$

We use  $\|x\|_1 = \sum_i |x_i|$ , and  $|I|$  denotes one-dimensional Lebesgue measure. Unless positive length is required explicitly, intervals  $I$  are used as their literal subsets and may be open, closed, half-open, empty, or singletons; they are not silently replaced by their closures. For a nondegenerate compact interval  $\Theta = [\theta_-, \theta_+]$ ,  $C^1$  regularity means ordinary  $C^1$  regularity on the interior with the first derivative extending continuously to  $\Theta$ ; derivatives at  $\theta_-$  and  $\theta_+$  are the corresponding one-sided endpoint derivatives. If  $\Theta$  is a singleton, the derivative value is part of the specified  $C^1$  presentation, equivalently of a compatible local  $C^1$  extension. Put

$$A = (2R)^N \kappa.$$

Let  $\mathcal{D}_{N,R,\kappa}$  be the class of Borel probability laws  $\mu$  on  $\mathbb{R}^N$  that have a full joint Lebesgue density  $f_\mu$  satisfying

$$f_\mu = 0 \quad \text{almost everywhere outside } [-R, R]^N, \quad \|f_\mu\|_{L^\infty(\mathbb{R}^N)} \leq \kappa.$$

This definition imposes no product structure or coordinate independence.

The deterministic family has a common one-dimensional triangular Pfaffian presentation

$$\begin{aligned} \eta'_j(\theta) &= P_j(\theta, \eta_1(\theta), \dots, \eta_j(\theta)), \quad 1 \leq j \leq q, \\ b(\theta) &= Q_0(\theta, \eta(\theta)), \quad F_i(\theta) = Q_i(\theta, \eta(\theta)), \quad 1 \leq i \leq N. \end{aligned}$$

In ambient parameter dimension  $p = 1$ , its descriptors are

$$M = \max_{1 \leq j \leq q} \deg P_j, \quad \Delta_{\text{rnd}} = \max_{1 \leq i \leq N} \deg Q_i, \quad \Delta_{\text{aff}} = \max_{0 \leq i \leq N} \deg Q_i.$$

When  $q = 0$  the chain is absent and  $M = 0$ . The offset  $b = Q_0$  contributes to  $\Delta_{\text{aff}}$  but is not a random coefficient.

## 1.2 Assumptions

**Assumption 1** (Shared one-dimensional Pfaffian regularity). *The functions  $b, F_1, \dots, F_N$  are  $C^1$  on  $\Theta$  and have the common triangular Pfaffian representation and the  $(q, M, \Delta_{\text{rnd}}, \Delta_{\text{aff}}, p = 1)$  degree convention above.*

**Assumption 2** (Primitive no-forced-root nondegeneracy). *For every  $\theta \in \Theta$ ,*

$$(b(\theta), F(\theta)) \neq (0, 0).$$

*Thus no parameter value is a root for every coefficient vector. Points at which  $F(\theta) = 0$  and  $b(\theta) \neq 0$  remain admissible and root-free.*

**Assumption 3** (Arbitrarily correlated bounded joint density). *The coefficient law is any  $\mu \in \mathcal{D}_{N,R,\kappa}$ . The bound is on the full joint density and does not impose independence.*

## 1.3 Objective and Scope

Our objective is an unconditional affine coordinate-pivot sweep theorem. It gives fixed-family finiteness of a static pivot-conditioning functional, an ordinary-probability root bound uniform over admissible laws and positive-length intervals, and two quantitative checks. The first check retains the exact  $1/\delta$  scale in a rescaled two-feature family. The second keeps the leading coefficient of a monic polynomial deterministic and recovers the exact coefficient for arbitrary correlated laws on its lower coefficients. No result below asserts a polynomial general-instance bound for the conditioning functional in  $q, M, \Delta_{\text{rnd}}, \Delta_{\text{aff}}$  or in other general Pfaffian presentation data.

## 2 Preliminaries

The root-feasible subset of the fixed parameter interval is

$$K_R = \{\theta \in \Theta : |b(\theta)| \leq R \|F(\theta)\|_1\}.$$

For coordinate  $j$ , let  $U_j = \{\theta \in \Theta : F_j(\theta) \neq 0\}$  and, on  $U_j$ , set

$$V_j(\theta) = \left| \left( \frac{b}{F_j} \right)'(\theta) \right| + R \sum_{i \neq j} \left| \left( \frac{F_i}{F_j} \right)'(\theta) \right|.$$

For minimization, extend  $V_j$  by  $+\infty$  on  $\Theta \setminus U_j$ . The static coordinate-pivot conditioning functional used in the theorem is

$$\Gamma_{\text{piv}}(b, F; R) = \begin{cases} \sup_{\theta \in K_R} \min_{1 \leq j \leq N} V_j(\theta), & K_R \neq \emptyset, \\ 0, & K_R = \emptyset. \end{cases}$$

It is defined before the coefficient vector is sampled.

Two setting-derived families will be used. For  $0 < \delta \leq 1$ , on  $\Theta = [-1, 1]$  and with  $R = 1$ , define

$$b_\delta(\theta) = 0, \quad F_\delta(\theta) = \left( 1, \frac{\theta}{\delta} \right).$$

For an integer  $d \geq 1$ , define the exact monic family

$$b_d(\theta) = \theta^d, \quad F_d(\theta) = (1, \theta, \dots, \theta^{d-1}), \quad p_\alpha(\theta) = \theta^d + \sum_{k=0}^{d-1} \alpha_k \theta^k.$$

Only  $\alpha = (\alpha_0, \dots, \alpha_{d-1})$  is random. For this family  $q = M = 0$ ,  $N = d$ ,  $\Delta_{\text{rnd}} = d - 1$ ,  $\Delta_{\text{aff}} = d$ , and  $A = (2R)^d \kappa$ .

### 3 Main Theorem

**Theorem 3.1** (Affine coordinate-pivot sweep and exact specializations). *Fix  $N \geq 1$ ,  $R > 0$ ,  $\kappa \in (0, \infty)$ , a compact interval  $\Theta \subset \mathbb{R}$ , and a deterministic affine family  $(b, F)$ . Under Assumptions 1 and 2,*

$$0 \leq \Gamma_{\text{piv}}(b, F; R) < \infty.$$

*Under Assumption 3 as well, every  $\mu \in \mathcal{D}_{N, R, \kappa}$  and every interval  $I \subseteq \Theta$  with  $|I| > 0$  satisfy*

$$\Pr_{\alpha \sim \mu} [\exists \theta \in I : b(\theta) + \langle \alpha, F(\theta) \rangle = 0] \leq \kappa (2R)^{N-1} \Gamma_{\text{piv}}(b, F; R) |I| = \frac{A \Gamma_{\text{piv}}(b, F; R)}{2R} |I|. \quad (\text{T1})$$

*With the interval supremum taken before the law supremum and with  $\sup \emptyset = -\infty$ ,*

$$\sup_{\mu \in \mathcal{D}_{N, R, \kappa}} \sup_{\substack{I \subseteq \Theta, \text{ } I \text{ an interval} \\ |I| > 0}} \frac{\Pr_{\alpha \sim \mu} [\exists \theta \in I : b(\theta) + \langle \alpha, F(\theta) \rangle = 0]}{|I|} \leq \frac{A \Gamma_{\text{piv}}(b, F; R)}{2R}. \quad (\text{T2})$$

*The left side of (T2) is a finite nonnegative real precisely in the branch in which both index classes are nonempty, namely  $A \geq 1$  and  $|\Theta| > 0$ . If  $A < 1$  or  $|\Theta| = 0$ , it is literally  $-\infty$ ; in that branch (T2) is a vacuous extended-real inequality and is not a finite nonnegative capacity statement.*

*For every  $0 < \delta \leq 1$ , including  $\delta = 1$ , the scale-stress family satisfies the exact identity*

$$\Gamma_{\text{piv}}(b_\delta, F_\delta; 1) = \frac{1}{\delta}. \quad (\text{T3})$$

*For every integer  $d \geq 1$ , every  $R > 0$ , and every compact restriction of the exact monic family,*

$$\Gamma_{\text{piv}}(b_d, F_d; R) \leq d + \frac{Rd(d-1)}{2}. \quad (\text{T4})$$

*Consequently, if  $\mu \in \mathcal{D}_{d, R, \kappa}$  is any full joint law of the  $d$  lower coefficients, with arbitrary correlation, then every bounded interval  $I \subset \mathbb{R}$ , including  $|I| = 0$ , satisfies*

$$\Pr_{\alpha \sim \mu} [\exists \theta \in I : p_\alpha(\theta) = 0] \leq \kappa (2R)^{d-1} \left( d + \frac{Rd(d-1)}{2} \right) |I|. \quad (\text{T5})$$

*The leading coefficient one in (T5) is deterministic and is not part of the random vector.*

*All displayed rates have no hidden constants. Their exposed variables are  $N, R, \kappa, A, \Gamma_{\text{piv}}, |I|$  and, in the specializations,  $\delta$  or  $d$ ; the Pfaffian descriptors are fixed family descriptors and do not enter as unshown factors. The probabilities are ordinary probabilities for each fixed law, the interval statements are static and uniform over their stated interval classes, and there is no confidence or horizon parameter. The norm and measure conventions are the  $\ell_1$  feasibility test defining  $K_R$ , scalar absolute coordinate-ratio variation, one-dimensional interval length, and full-dimensional Lebesgue density. No independence or probability conversion is implicit. The theorem does not bound  $\Gamma_{\text{piv}}$  polynomially in general Pfaffian presentation data.*

### 4 Proof Sketch

Compactness of  $K_R$  and the no-forced-root condition give a positive available-coordinate margin on the feasible set. Quotient differentiation then proves fixed-family finiteness of  $\Gamma_{\text{piv}}$ . The extended pivot speeds are Borel, so the least minimizing coordinate partitions  $K_R$  into disjoint Borel cells. On each cell, solving for the selected coefficient produces a graph in the original coefficient coordinates whose  $\theta$ -velocity is bounded by the selected pivot speed.

The selected denominators are exhausted by the sets  $|F_j| \geq 1/m$ . Each exhausted graph is localized into countably many Lipschitz pieces. The equal-dimensional area formula bounds its image by the integral of the exact graph Jacobian. Descriptive-set results ensure that the chart images and the full root event are Lebesgue measurable. Monotone exhaustion recovers every nonzero selected pivot, and disjointness of the parameter cells converts the chart sum into the single length  $|I \cap K_R|$ . Integrating the resulting coefficient-volume estimate against the capped full joint density gives (T1). Separate sharp nonemptiness tests for the law and interval classes give the literal branch semantics of (T2).

For the scale-stress family, both coordinate speeds are computed exactly; the constant pivot gives a global upper bound  $1/\delta$  and the value at  $\theta = 0$  gives equality. For monic polynomials, the constant coordinate controls  $|\theta| \leq 1$ , while the degree- $(d - 1)$  coordinate controls  $|\theta| > 1$ . Their explicit quotient derivatives give (T4). Applying (T1) on the compact closure of a positive-length bounded interval gives the exact coefficient in (T5). Empty and singleton intervals are handled by direct affine-hyperplane nullity under the full joint density.

## References

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## A Proof Details

### A.1 Root Feasibility and Finite Pivot Conditioning

**Lemma A.1** (Compact feasibility of cube-supported roots). *Under Assumption 1, for  $N \geq 1$ ,  $R > 0$ , and compact interval  $\Theta$ , the set*

$$K_R = \left\{ \theta \in \Theta : |b(\theta)| \leq R \sum_{i=1}^N |F_i(\theta)| \right\}$$

*is compact. For every  $\alpha \in [-R, R]^N$  and  $\theta \in \Theta$ ,  $\phi_\alpha(\theta) = 0$  implies  $\theta \in K_R$ . If  $K_R = \emptyset$ , then*

$$\{\alpha \in [-R, R]^N : \exists \theta \in \Theta, \phi_\alpha(\theta) = 0\} = \emptyset, \quad \Gamma_{\text{piv}}(b, F; R) = 0.$$

*Proof.* Assumption 1 makes  $b, F_1, \dots, F_N$  continuous. Hence

$$\theta \mapsto |b(\theta)| - R \sum_{i=1}^N |F_i(\theta)|$$

is continuous, and  $K_R$  is its inverse image of  $(-\infty, 0]$ . It is closed relative to compact  $\Theta$ , and therefore compact. This relative topology argument retains both endpoints of  $\Theta$ .

If  $\alpha \in [-R, R]^N$  and  $\phi_\alpha(\theta) = 0$ , then  $b(\theta) = -\sum_i \alpha_i F_i(\theta)$ , so the finite triangle inequality gives

$$\begin{aligned} |b(\theta)| &= \left| \sum_{i=1}^N \alpha_i F_i(\theta) \right| \\ &\leq \sum_{i=1}^N |\alpha_i| |F_i(\theta)| \\ &\leq R \sum_{i=1}^N |F_i(\theta)| = R \|F(\theta)\|_1. \end{aligned}$$

Thus  $\theta \in K_R$ . When  $K_R$  is empty, this implication rules out every cube-supported root. The second branch in the definition of  $\Gamma_{\text{piv}}$  then gives zero; no empty-supremum convention is being used here.  $\square$

**Lemma A.2** (Positive available-pivot margin). *Under Assumptions 1 and 2 and Lemma A.1, if  $K_R \neq \emptyset$ , then  $F(\theta) \neq 0$  for every  $\theta \in K_R$  and*

$$\rho := \min_{\theta \in K_R} \max_{1 \leq j \leq N} |F_j(\theta)| > 0.$$

*At each  $\theta \in K_R$ , every index attaining the displayed coordinate maximum is a legal pivot and has  $|F_j(\theta)| \geq \rho$ .*

*Proof.* If  $\theta \in K_R$  and  $F(\theta) = 0$ , feasibility gives

$$|b(\theta)| \leq R \|F(\theta)\|_1 = 0.$$

Then  $(b(\theta), F(\theta)) = (0, 0)$ , contrary to Assumption 2. This argument is confined to  $K_R$ ; a point outside  $K_R$  with  $F = 0$  and  $b \neq 0$  remains admissible and root-free.

The function

$$g(\theta) = \max_{1 \leq j \leq N} |F_j(\theta)|$$

is continuous as a finite maximum of continuous functions, and it is strictly positive on  $K_R$ . Lemma A.1 makes  $K_R$  nonempty and compact in the present branch, so  $g$  attains a minimum at some  $\theta_* \in K_R$ . That minimum is  $\rho = g(\theta_*) > 0$ . The finite coordinate set has a maximizer at each point, and every maximizer has magnitude  $g(\theta) \geq \rho$ .

Coordinates may vanish and maximizers may tie; at least one maximizer is nonzero. The choice here is pointwise, with no assertion of a continuous or measurable global index. When  $N = 1$ , the sole coordinate is the maximizer and is nonzero on  $K_R$ . Compact-set endpoints are included and no transversality condition is used.  $\square$

**Proposition A.1** (Crude fixed-family pivot-variation bound). *Under Assumptions 1 and 2 and Lemmas A.1 and A.2, suppose  $K_R \neq \emptyset$  and define*

$$B_0 := \max_{\theta \in \Theta} \max\{|b(\theta)|, |F_1(\theta)|, \dots, |F_N(\theta)|\},$$

$$B_1 := \max_{\theta \in \Theta} \max\{|b'(\theta)|, |F'_1(\theta)|, \dots, |F'_N(\theta)|\}.$$

Then  $B_0, B_1 < \infty$ , and for every  $\theta \in K_R$ ,

$$\min_{1 \leq k \leq N} V_k(\theta) \leq 2B_0B_1\rho^{-2}(1 + R(N - 1)).$$

Consequently,

$$\Gamma_{\text{piv}}(b, F; R) \leq 2B_0B_1\rho^{-2}(1 + R(N - 1)) < \infty.$$

*Proof.* The functions  $b, F_i, b', F'_i$  are continuous on compact  $\Theta$ , so the two displayed maxima exist and are finite. Nonempty feasibility also gives  $B_0 \geq \rho > 0$ , although we never divide by  $B_0$ ;  $B_1$  need not be positive.

Fix  $\theta \in K_R$  and choose an index  $j$  attaining  $\max_k |F_k(\theta)|$ . Lemma A.2 gives  $|F_j(\theta)| \geq \rho > 0$ . The quotient rule and the two-term triangle inequality give

$$\begin{aligned} \left| \left( \frac{b}{F_j} \right)'(\theta) \right| &= \frac{|b'(\theta)F_j(\theta) - b(\theta)F'_j(\theta)|}{|F_j(\theta)|^2} \\ &\leq \frac{|b'(\theta)||F_j(\theta)| + |b(\theta)||F'_j(\theta)|}{|F_j(\theta)|^2} \\ &\leq \frac{B_1B_0 + B_0B_1}{\rho^2} = \frac{2B_0B_1}{\rho^2}. \end{aligned}$$

For each  $i \neq j$ , the separate quotient calculation is

$$\begin{aligned} \left| \left( \frac{F_i}{F_j} \right)'(\theta) \right| &= \frac{|F'_i(\theta)F_j(\theta) - F_i(\theta)F'_j(\theta)|}{|F_j(\theta)|^2} \\ &\leq \frac{|F'_i(\theta)||F_j(\theta)| + |F_i(\theta)||F'_j(\theta)|}{|F_j(\theta)|^2} \\ &\leq \frac{2B_0B_1}{\rho^2}. \end{aligned}$$

There is one offset ratio and exactly  $N - 1$  feature ratios. Therefore

$$\begin{aligned} V_j(\theta) &= \left| \left( \frac{b}{F_j} \right)'(\theta) \right| + R \sum_{i \neq j} \left| \left( \frac{F_i}{F_j} \right)'(\theta) \right| \\ &\leq \frac{2B_0B_1}{\rho^2} + R(N - 1) \frac{2B_0B_1}{\rho^2} \\ &= 2B_0B_1\rho^{-2}(1 + R(N - 1)). \end{aligned}$$

Although zero pivots have extended speed  $+\infty$ , the selected  $j$  is nonzero; hence  $\min_k V_k(\theta) \leq V_j(\theta) < \infty$ . Taking the supremum over nonempty  $K_R$  proves the result.

If an unselected  $F_i(\theta)$  vanishes, its numerator term  $F_i F'_j$  is zero while  $F'_i F_j$  is covered by the displayed bound. If  $b(\theta) = 0$ , its corresponding term vanishes; if  $b \equiv 0$ , then  $b' \equiv 0$  and the offset-ratio derivative is zero. When  $B_1 = 0$ , every displayed derivative is zero. For  $N = 1$  the feature-ratio sum is empty and  $1 + R(N - 1) = 1$ . The  $C^1$  convention includes endpoints. This certificate is fixed-family: it gives no bound for  $B_0, B_1$ , or  $\rho^{-1}$  in terms of  $q, M, \Delta_{\text{rnd}}, \Delta_{\text{aff}}$  or other general presentation data.  $\square$

**Proposition A.2** (Root feasibility and finite pivot conditioning). *Under Assumptions 1 and 2, every root generated by  $\alpha \in [-R, R]^N$  lies in compact  $K_R$ , a legal pivot is available at every feasible point, and*

$$0 \leq \Gamma_{\text{piv}}(b, F; R) < \infty.$$

*If  $K_R = \emptyset$ , the supported-root event is empty and  $\Gamma_{\text{piv}} = 0$ ; otherwise the bound in Proposition A.1 holds.*

*Proof.* Lemma A.1 gives compactness, the supported-root implication, and the complete empty branch. In the nonempty branch, Lemma A.2 gives  $\rho > 0$  and a nonzero largest-coordinate pivot at each point. Proposition A.1 bounds all quotient terms for such a pivot and then takes the defining supremum. These two branches prove every assertion. The quantities  $B_0, B_1, \rho$  witness finiteness but are not additional public rate factors.  $\square$

## A.2 Measurable Adaptive Coordinate Charts

**Lemma A.3** (Borel measurability of extended pivot speeds). *Under Assumption 1 and Proposition A.2, every extension  $V_j : \Theta \rightarrow [0, +\infty]$  is Borel. If  $K_R \neq \emptyset$ , then for every  $\theta \in K_R$ ,*

$$\min_{1 \leq i \leq N} V_i(\theta) \leq \Gamma_{\text{piv}}(b, F; R) < \infty.$$

*For empty  $K_R$  the pointwise statement is vacuous and  $\Gamma_{\text{piv}} = 0$ .*

*Proof.* Fix  $j$ . Continuity of  $F_j$  makes  $U_j = \{\theta : F_j(\theta) \neq 0\}$  relatively open and Borel. On  $U_j$ ,

$$\left(\frac{b}{F_j}\right)' = \frac{b'F_j - bF'_j}{F_j^2}, \quad \left(\frac{F_i}{F_j}\right)' = \frac{F'_i F_j - F_i F'_j}{F_j^2} \quad (i \neq j).$$

All numerator factors are continuous and the denominator is nonzero there, so

$$W_j(\theta) = \left| \left(\frac{b}{F_j}\right)'(\theta) \right| + R \sum_{i \neq j} \left| \left(\frac{F_i}{F_j}\right)'(\theta) \right|$$

is finite and continuous on  $U_j$ , including in the relative topology at endpoints. Since  $V_j = W_j$  on  $U_j$  and  $V_j = +\infty$  elsewhere, for every finite  $a$ ,

$$\{\theta : V_j(\theta) < a\} = \{\theta \in U_j : W_j(\theta) < a\},$$

which is Borel. Finite strict sublevel sets generate the Borel structure on  $[0, +\infty]$ , so  $V_j$  is Borel. The same argument proves

$$V_j(\theta) < +\infty \iff F_j(\theta) \neq 0.$$

For nonempty  $K_R$ , Proposition A.2 provides a finite ordinary speed at each feasible point. A finite list with one finite entry has a finite attained minimum, and the definition of  $\Gamma_{\text{piv}}$  gives the displayed bound. The empty branch follows from the same proposition. No continuity across a zero of  $F_j$  is asserted.  $\square$



**Proposition A.3** (Borel lexicographic pivot partition). *Under Lemma A.3 and Lemma A.1, define for  $\theta \in K_R$*

$$j_*(\theta) = \min_{1 \leq i \leq N} \arg \min V_i(\theta).$$

*Then  $j_* : K_R \rightarrow \{1, \dots, N\}$  is Borel. Its fibers  $E_j = \{\theta \in K_R : j_*(\theta) = j\}$  obey*

$$E_j = K_R \cap \{V_j < +\infty\} \cap \bigcap_{i < j} \{V_j < V_i\} \cap \bigcap_{i > j} \{V_j \leq V_i\}. \quad (\text{A1})$$

*They are Borel, pairwise disjoint, and cover  $K_R$ . On  $E_j$ ,*

$$V_j(\theta) = \min_i V_i(\theta) < +\infty, \quad F_j(\theta) \neq 0.$$

*These conclusions also hold for empty  $K_R$ , with empty fibers.*

*Proof.* For extended-real Borel functions  $f, g$ ,

$$\{f < g\} = \bigcup_{r \in \mathbb{Q}} (\{f < r\} \cap \{g > r\}), \quad \{f \leq g\} = \Theta \setminus \{g < f\}. \quad (\text{A2})$$

The first equality follows by rational separation when  $g$  is finite and, when  $g = +\infty$ , by choosing a finite rational larger than  $f$ . Thus all comparisons in (A1) are Borel, as are  $K_R$  and  $\{V_j < +\infty\}$ .

At each  $\theta \in K_R$ , Lemma A.3 gives a finite attained minimum. If  $j$  is its least attaining index, then  $V_j < V_i$  for  $i < j$  and  $V_j \leq V_i$  for  $i > j$ , placing the point in  $E_j$ . Conversely, (A1) makes  $j$  the least minimizer. This proves coverage and uniqueness, hence disjointness. Each singleton fiber of  $j_*$  is Borel; inverse images of subsets of the finite discrete target are finite unions of these fibers, so  $j_*$  is Borel. Finiteness of  $V_j$  on its fiber and Lemma A.3 give  $F_j \neq 0$ .

When  $N = 1$ , both comparison intersections are empty,  $E_1 = K_R$ , and the selector is one on its domain. Empty  $K_R$  gives the unique empty map. Zeros in unselected coordinates are unrestricted.  $\square$

**Lemma A.4** (Exact denominator exhaustion). *Under Assumption 1 and Proposition A.3, for  $m \geq 1$  define*

$$E_{j,m} = E_j \cap \{\theta \in \Theta : |F_j(\theta)| \geq 1/m\}.$$

*Then every  $E_{j,m}$  is Borel and*

$$E_{j,m} \subseteq E_{j,m+1}, \quad \bigcup_{m=1}^{\infty} E_{j,m} = E_j. \quad (\text{A3})$$

*Every selected point enters at a finite level.*

*Proof.* Continuity of  $F_j$  makes  $\{|F_j| \geq 1/m\}$  relatively closed, hence Borel; intersecting with Borel  $E_j$  preserves Borelness. Since  $1/(m+1) \leq 1/m$ , the sets increase. Their union is contained in  $E_j$ . For the reverse inclusion, if  $\theta \in E_j$  then  $x = |F_j(\theta)| > 0$  by Proposition A.3. There is a finite integer  $m$  with  $1/m \leq x$ , so  $\theta \in E_{j,m}$ . The weak threshold retains equality points. An arbitrarily small positive pivot may enter late but leaves no remainder; empty cells satisfy the same identity.  $\square$

**Proposition A.4** (Root chart in the original coefficient coordinates). *Under Assumption 1 and Proposition A.3, fix  $j$  and set  $J_j = \{1, \dots, N\} \setminus \{j\}$ , ordered increasingly. For  $\beta = (\beta_i)_{i \in J_j}$  define*

$$T_j(\theta, \beta) = -\frac{b(\theta) + \sum_{i \in J_j} \beta_i F_i(\theta)}{F_j(\theta)}. \quad (\text{A4})$$

This function is well-defined and jointly Borel on  $E_j \times [-R, R]^{J_j}$ . If

$$\alpha_i^{(j)}(\theta, \beta) = \begin{cases} T_j(\theta, \beta), & i = j, \\ \beta_i, & i \in J_j, \end{cases}$$

then

$$b(\theta) + \langle \alpha^{(j)}(\theta, \beta), F(\theta) \rangle = 0. \quad (\text{A5})$$

The conclusion does not assert  $T_j \in [-R, R]$ .

*Proof.* The selected cell lies in  $U_j$ . On  $U_j \times \mathbb{R}^{J_j}$ , rewrite (A4) as

$$-\frac{b(\theta)}{F_j(\theta)} - \sum_{i \in J_j} \beta_i \frac{F_i(\theta)}{F_j(\theta)}.$$

The coefficient ratios are continuous on  $U_j$  and the expression is finite affine in  $\beta$ , hence jointly continuous. Its restriction to the stated Borel domain is Borel. Direct substitution gives

$$\begin{aligned} b + \langle \alpha^{(j)}, F \rangle &= b + T_j F_j + \sum_{i \in J_j} \beta_i F_i \\ &= b - \left( b + \sum_{i \in J_j} \beta_i F_i \right) + \sum_{i \in J_j} \beta_i F_i = 0. \end{aligned}$$

Thus the residual vanishes in the original coefficient vector; there is no weighted or transformed target. Unselected features may vanish. If  $N = 1$ ,  $J_1 = \emptyset$ , the beta space is the singleton  $\mathbb{R}^0$ , both sums are empty, and (A5) reads  $b + (-b/F_1)F_1 = 0$ . Empty cells give empty domains, and the relative  $C^1$  convention includes interval endpoints.  $\square$

**Proposition A.5** (Selected-chart velocity). *Under Assumption 1, Proposition A.2, Proposition A.3, and Proposition A.4, the following holds for every  $\theta \in E_j$  and every  $\beta \in [-R, R]^{J_j}$ :*

$$\partial_\theta T_j(\theta, \beta) = - \left( \frac{b}{F_j} \right)'(\theta) - \sum_{i \in J_j} \beta_i \left( \frac{F_i}{F_j} \right)'(\theta), \quad (\text{A6})$$

and

$$|\partial_\theta T_j(\theta, \beta)| \leq V_j(\theta) = \min_{1 \leq i \leq N} V_i(\theta) \leq \Gamma_{\text{piv}}(b, F; R) < \infty. \quad (\text{A7})$$

The same conclusions hold on every exhausted domain  $E_{j,m} \times [-R, R]^{J_j}$ .

*Proof.* The selected denominator is nonzero. On  $U_j$ , the quotient identities used in Lemma A.3 hold. Differentiating the finite affine expression (A4) at fixed  $\beta$  gives (A6) without interchanging any limit or infinite sum. At an endpoint this is the derivative in the stated  $C^1$  convention.

Using  $|\beta_i| \leq R$  separately for every nonpivot coordinate,

$$\begin{aligned} |\partial_\theta T_j| &\leq \left| \left( \frac{b}{F_j} \right)' \right| + \sum_{i \in J_j} |\beta_i| \left| \left( \frac{F_i}{F_j} \right)' \right| \\ &\leq \left| \left( \frac{b}{F_j} \right)' \right| + R \sum_{i \in J_j} \left| \left( \frac{F_i}{F_j} \right)' \right| = V_j. \end{aligned}$$

Cell membership gives  $V_j = \min_i V_i$ , and  $E_j \subseteq K_R$  together with Proposition A.2 gives the last bound in (A7). There is one literal factor  $R$  per nonpivot term and no sum over chart indices. Restriction to  $E_{j,m}$  changes no formula and introduces no  $m$ -dependent constant. For  $N = 1$  the sum is empty and  $|(b/F_1)'| = V_1$ . If  $K_R$  is empty, the domain is empty and  $\Gamma_{\text{piv}} = 0$ .  $\square$

**Proposition A.6** (Measurable adaptive coordinate charts). *Under Assumptions 1 and 2, the least finite-speed selector gives a Borel disjoint partition  $K_R = \bigsqcup_{j=1}^N E_j$  with nonzero selected denominators, the Borel sets  $E_{j,m}$  increase exactly to  $E_j$ , and the original-coordinate chart (A4) satisfies (A5) and (A7). These conclusions include ties, zero unselected features, endpoints, empty  $K_R$ , and  $N = 1$ .*

*Proof.* Lemma A.3 provides extended-real Borel speeds and a finite minimum at every feasible point. Proposition A.3 applies the strict-before and weak-after comparisons in (A1), resolving all ties by the least index. Lemma A.4 proves both inclusions in (A3) and finite-level entry. Finally, Proposition A.4 gives the same-object root identity, while Proposition A.5 differentiates that expression and proves the literal  $R$ -weighted bound. No probability law, area estimate, or chart-count factor is used here.  $\square$

### A.3 Measurable Swept Coefficient Volume

**Lemma A.5** (Analytic exhausted charts and root event). *Under Assumption 1 and Proposition A.6, let  $I \subseteq \Theta$  be any interval. For*

$$J_j = \{1, \dots, N\} \setminus \{j\}, \quad B_j = [-R, R]^{J_j},$$

*identify  $\mathbb{R}^{J_j}$  with  $\mathbb{R}^{N-1}$  in increasing original coordinate order and define*

$$\Psi_j(\theta, \beta)_i = \begin{cases} T_j(\theta, \beta), & i = j, \\ \beta_i, & i \in J_j. \end{cases} \quad (\text{A8})$$

*For  $m \geq 1$ , set*

$$D_{j,m} = \{(\theta, \beta) : \theta \in I \cap E_{j,m}, \beta \in B_j, |T_j(\theta, \beta)| \leq R\}. \quad (\text{A9})$$

*Then  $D_{j,m}$  is Borel in  $\mathbb{R} \times \mathbb{R}^{J_j} \cong \mathbb{R}^N$ , and  $\Psi_j$  is the restriction of a  $C^1$ , locally Lipschitz map on an ambient open nonzero-pivot chart. Every  $\Psi_j(D_{j,m})$  is analytic and Lebesgue measurable. The set*

$$S_I = \{\alpha \in [-R, R]^N : \exists \theta \in I, b(\theta) + \langle \alpha, F(\theta) \rangle = 0\} \quad (\text{A10})$$

*is also analytic and Lebesgue measurable. This includes open, closed, half-open, empty, and singleton intervals; endpoints of  $\Theta$ ; cube faces and corners; empty chart domains; and  $N = 1$ .*

*Proof.* Write  $\Theta = [\theta_-, \theta_+]$ . When  $\Theta$  is nondegenerate, extend each  $g \in \{b, F_1, \dots, F_N\}$  to  $\mathbb{R}$  by

$$\tilde{g}(x) = \begin{cases} g(\theta_-) + g'(\theta_-)(x - \theta_-), & x < \theta_-, \\ g(x), & \theta_- \leq x \leq \theta_+, \\ g(\theta_+) + g'(\theta_+)(x - \theta_+), & x > \theta_+. \end{cases} \quad (\text{A11})$$

Values and first derivatives agree at both endpoints, so  $\tilde{g}$  is  $C^1$ . If  $\Theta$  is a singleton, use the single affine expression based at that point and its derivative in the endpoint-inclusive  $C^1$  convention. These extensions agree with the original values and derivatives throughout  $\Theta$ .

For each  $j$ , let

$$O_j = \{(\theta, \beta) \in \mathbb{R} \times \mathbb{R}^{J_j} : \tilde{F}_j(\theta) \neq 0\}. \quad (\text{A12})$$

This set is open. Replacing the functions in (A4) by their extensions defines a  $C^1$  function  $\tilde{T}_j$  on  $O_j$ , and insertion as in (A8) defines a  $C^1$  map  $\tilde{\Psi}_j : O_j \rightarrow \mathbb{R}^N$ . On the selected original chart these maps equal  $T_j$  and  $\Psi_j$ . Consequently, even when  $\theta$  is an endpoint, every point of  $D_{j,m}$  has an ambient open  $C^1$  neighborhood. This construction imposes no uniform lower bound on a prescribed pivot.

Every interval in  $\mathbb{R}$  is Borel, Lemma A.4 makes  $E_{j,m}$  Borel, and  $B_j$  is closed. Proposition A.3 gives  $E_{j,m} \subseteq E_j \subseteq \{F_j \neq 0\}$ , so the base product in (A9) is a Borel subset of  $O_j$ . The final condition in (A9) is the inverse image of  $[-R, R]$  under the continuous ambient map  $\tilde{T}_j$ . Thus  $D_{j,m}$  is Borel and the restricted map is Borel.

Kechris's Proposition 14.4 states that if  $X$  and  $Y$  are Polish spaces,  $f : X \rightarrow Y$  is Borel, and  $A \subseteq X$  is analytic, then  $f(A)$  is analytic (Kechris, 1995, Section 14, Proposition 14.4, p. 86). Here  $O_j$  is an open Polish subspace of  $\mathbb{R} \times \mathbb{R}^{J_j}$ , the Borel set  $D_{j,m} \subseteq O_j$  is analytic in  $O_j$ , and the global map  $\tilde{\Psi}_j : O_j \rightarrow \mathbb{R}^N$  is  $C^1$  and hence Borel. Therefore  $\Psi_j(D_{j,m}) = \tilde{\Psi}_j(D_{j,m})$  is analytic.

For the full event, define the incidence set

$$Z_I = \{(\theta, \alpha) \in I \times [-R, R]^N : b(\theta) + \langle \alpha, F(\theta) \rangle = 0\}. \quad (\text{A13})$$

The evaluation map is continuous and the containing product is Borel, so  $Z_I$  is Borel, and hence analytic, in the Polish space  $\Theta \times \mathbb{R}^N$ . The global coefficient projection  $\pi_\alpha : \Theta \times \mathbb{R}^N \rightarrow \mathbb{R}^N$  is continuous and satisfies  $\pi_\alpha(Z_I) = S_I$ . The same cited proposition therefore makes  $S_I$  analytic without relying on the graph-coverage identity proved below. These two applications supply analyticity only; universal and Lebesgue measurability are obtained separately next, and the coverage, Jacobian, and volume conclusions are proved later without attribution to Proposition 14.4.

Every set just shown analytic lies in  $C = [-R, R]^N$ : this is explicit in (A10), and (A9) puts the beta coordinates and the inserted  $T_j$  coordinate in  $C$ . Every analytic subset of a Polish space is universally measurable (Kechris, 1995, Section 21, Theorem 21.10, p. 155). Apply this to the Borel probability measure

$$\nu_C(B) = \frac{\lambda_N(B \cap C)}{(2R)^N}.$$

If  $A_0 \subseteq C$  is one of the analytic sets, universal measurability gives a Borel set  $B_0$  and a Borel  $\nu_C$ -null envelope  $M$  with  $A_0 \triangle B_0 \subseteq M$ . Since  $A_0 \subseteq C$ ,

$$A_0 \triangle (B_0 \cap C) \subseteq M \cap C, \quad \lambda_N(M \cap C) = 0.$$

Hence  $A_0$  is Lebesgue measurable in  $\mathbb{R}^N$ .

No endpoint or coefficient boundary was removed: the original interval, closed beta cube, and weak inequality  $|T_j| \leq R$  remain in (A9). For  $N = 1$ ,  $J_1 = \emptyset$  and  $\mathbb{R}^{J_1} = \mathbb{R}^0 = \{()\}$ ; the source is still one-dimensional and the same arguments apply. Empty domains, images, and events are analytic and measurable.  $\square$

**Proposition A.7** (Localized equal-dimensional area sweep). *Under Assumption 1, Propositions A.4 and A.5, and Lemma A.5, order the source coordinates as*

$$(\theta, \beta_1, \dots, \beta_{j-1}, \beta_{j+1}, \dots, \beta_N)$$

*and retain target order  $(\alpha_1, \dots, \alpha_N)$ . For every  $j$  and  $m \geq 1$ ,*

$$\int_{D_{j,m}} |\det D\Psi_j(x)| d\lambda_N(x) = \int_{\mathbb{R}^N} N(\Psi_j, D_{j,m}, a) d\lambda_N(a), \quad (\text{A14})$$

where

$$N(\Psi_j, D_{j,m}, a) = \#(D_{j,m} \cap \Psi_j^{-1}(a)) \in \{0, 1, 2, \dots\} \cup \{+\infty\}.$$

Moreover,

$$|\det D\Psi_j(\theta, \beta)| = |\partial_\theta T_j(\theta, \beta)|, \quad (\text{A15})$$

and

$$\lambda_N(\Psi_j(D_{j,m})) \leq \int_{D_{j,m}} |\partial_\theta T_j| d\lambda_N \leq (2R)^{N-1} \int_{I \cap E_{j,m}} V_j(\theta) d\theta. \quad (\text{A16})$$

No injectivity, finite-fiber, regular-value, transversality, or simple-root condition is imposed. For  $N = 1$ , (A15) is scalar and the zero-dimensional beta measure in (A16) is one.

*Proof.* On the open set  $O_j$  from Lemma A.5, every target component other than  $j$  is its corresponding beta coordinate and target component  $j$  is  $\tilde{T}_j$ . Move target row  $j$  to the first row, leaving all other target rows in their relative order. The row permutation has determinant  $(-1)^{j-1}$ , and the permuted derivative is

$$\begin{pmatrix} \partial_\theta \tilde{T}_j & \nabla_\beta \tilde{T}_j \\ 0 & I_{N-1} \end{pmatrix}.$$

It has determinant  $\partial_\theta \tilde{T}_j$ , and therefore in the original target order

$$\det D\tilde{\Psi}_j = (-1)^{j-1} \partial_\theta \tilde{T}_j.$$

Restriction to the original chart proves (A15). The beta derivatives occupy only the upper-right block and contribute no determinant factor.

We next reduce local Lipschitz regularity to the global interface of the area formula. Cover  $O_j$  by a countable family of open balls  $(\mathcal{B}_\ell)_{\ell \geq 1}$  with rational centers and radii and with  $\overline{\mathcal{B}_\ell}$  compactly contained in  $O_j$ . The derivative of  $\tilde{\Psi}_j$  is bounded on each compact closure, so integration of the derivative along line segments in the convex ball makes  $\tilde{\Psi}_j$  Lipschitz there. Define disjoint Borel layers

$$\mathcal{L}_\ell = \mathcal{B}_\ell \setminus \bigcup_{r < \ell} \mathcal{B}_r, \quad D_{j,m}^{(\ell)} = D_{j,m} \cap \mathcal{L}_\ell. \quad (\text{A17})$$

They partition  $D_{j,m}$ .

For a scalar  $L$ -Lipschitz coordinate  $u$  on  $\mathcal{B}_\ell$ , the formula

$$\hat{u}(x) = \inf_{z \in \mathcal{B}_\ell} \{u(z) + L|x - z|\} \quad (\text{A18})$$

defines a global  $L$ -Lipschitz extension agreeing with  $u$  on the ball. Applying (A18) coordinatewise gives a global Lipschitz map  $\hat{\Psi}_{j,\ell} : \mathbb{R}^N \rightarrow \mathbb{R}^N$  that agrees with  $\tilde{\Psi}_j$  throughout  $\mathcal{B}_\ell$ , and hence has the same derivative on  $D_{j,m}^{(\ell)}$ .

The area formula with extended multiplicity, applied to this global map and the Borel domain piece, gives (Federer, 1969, Section 3.2.3, p. 243)

$$\int_{D_{j,m}^{(\ell)}} |\det D\Psi_j| d\lambda_N = \int_{\mathbb{R}^N} \#(D_{j,m}^{(\ell)} \cap \Psi_j^{-1}(a)) d\lambda_N(a). \quad (\text{A19})$$

Here source and target both have Euclidean dimension  $N$ ; Federer's normalized  $N$ -dimensional Hausdorff measure is  $\lambda_N$  in this setting. Both sides of (A19) are nonnegative extended integrals. Summing over  $\ell$ , exchanging the sum with the nonnegative integral, and using disjointness of (A17) yields

$$\begin{aligned} \int_{D_{j,m}} |\det D\Psi_j| d\lambda_N &= \int_{\mathbb{R}^N} \sum_{\ell \geq 1} \#(D_{j,m}^{(\ell)} \cap \Psi_j^{-1}(a)) d\lambda_N(a) \\ &= \int_{\mathbb{R}^N} \#(D_{j,m} \cap \Psi_j^{-1}(a)) d\lambda_N(a), \end{aligned}$$

which proves (A14) with no localization count or constant.

Lemma A.5 makes the image measurable. The multiplicity in (A14) is at least one on that image, even when it is infinite. Thus (A14) and (A15) give the first inequality in (A16). For the second, nonnegative product integration and Proposition A.5 give

$$\begin{aligned}
\int_{D_{j,m}} |\partial_\theta T_j| d\lambda_N &= \int_{I \cap E_{j,m}} \int_{B_j} \mathbf{1}_{\{|T_j(\theta, \beta)| \leq R\}} |\partial_\theta T_j(\theta, \beta)| d\lambda_{N-1}(\beta) d\theta \\
&\leq \int_{I \cap E_{j,m}} \int_{B_j} V_j(\theta) d\lambda_{N-1}(\beta) d\theta \\
&= \lambda_{N-1}(B_j) \int_{I \cap E_{j,m}} V_j(\theta) d\theta \\
&= (2R)^{N-1} \int_{I \cap E_{j,m}} V_j(\theta) d\theta.
\end{aligned} \tag{A20}$$

All faces of  $B_j$  and the two faces  $T_j = \pm R$  remain in the domain. For  $N = 1$ ,  $B_1 = [-R, R]^0 = \{()\}$  and  $\lambda_0(B_1) = 1 = (2R)^0$ .

For completeness, let  $a = \Psi_j(\theta, \beta)$ . Differentiating the root identity at fixed  $\beta$  gives

$$\partial_\theta T_j(\theta, \beta) = -\frac{\phi'_a(\theta)}{F_j(\theta)}. \tag{A21}$$

Tangent and differentiable multiple roots are therefore critical preimages. The Borel critical subset  $D_{j,m} \cap \{\partial_\theta T_j = 0\}$  has zero Jacobian, and applying the already established image inequality to that subset makes its image  $N$ -dimensionally null. If  $\phi_a$  vanishes identically on an interval, the represented fiber may be uncountable. Finite, countably infinite, and interval-sized fibers are all permitted by the extended multiplicity in (A14). Distinct or repeated roots add preimages; the proof never uses an inverse change of variables.  $\square$

**Lemma A.6** (Original-coordinate exhausted coverage). *Under Lemma A.1, Proposition A.3, Lemma A.4, and Proposition A.4, for every interval  $I \subseteq \Theta$  and every  $j$ ,*

$$\begin{aligned}
D_{j,m} &\subseteq D_{j,m+1}, \\
\Psi_j(D_{j,m}) &\subseteq \Psi_j(D_{j,m+1}),
\end{aligned} \tag{A22}$$

and

$$S_I = \bigcup_{j=1}^N \bigcup_{m=1}^{\infty} \Psi_j(D_{j,m}) \tag{A23}$$

as literal equality in the original coefficient cube. This includes interval and cube boundaries, least-index ties, arbitrarily small nonzero selected pivots, any number of roots, empty  $I$ , singleton  $I$ , empty  $K_R$ , empty cells, and  $N = 1$ .

*Proof.* Only the condition  $\theta \in E_{j,m}$  in (A9) depends on  $m$ . Lemma A.4 gives  $E_{j,m} \subseteq E_{j,m+1}$ ;  $I, B_j, T_j$ , and the weak range condition remain fixed. This proves domain nesting, and a fixed map sends nested domains to nested images.

Take  $a \in S_I$ . A witness  $\theta \in I$  satisfies  $\phi_a(\theta) = 0$ . Lemma A.1 gives  $\theta \in K_R$ , and Proposition A.3 assigns it to exactly one least-index cell  $E_j$  with  $F_j(\theta) \neq 0$ . Set  $\beta = a_{-j} \in B_j$  in increasing original-coordinate order. The root equation gives

$$a_j = -\frac{b(\theta) + \sum_{i \neq j} a_i F_i(\theta)}{F_j(\theta)} = T_j(\theta, \beta). \tag{A24}$$

Lemma A.4 puts this individual nonzero pivot in some finite  $E_{j,m}$ . Since  $|a_j| \leq R$ , the pair belongs to  $D_{j,m}$ , and (A8) yields  $\Psi_j(\theta, \beta) = a$ . This proves one inclusion in (A23).

Conversely, suppose  $a = \Psi_j(\theta, \beta)$  with  $(\theta, \beta) \in D_{j,m}$ . The beta coordinates lie in the closed cube and  $|T_j| \leq R$ , so  $a \in [-R, R]^N$  coordinate by coordinate, including faces and corners. Also  $\theta \in I \cap E_{j,m}$ , and Proposition A.4 gives  $b(\theta) + \langle a, F(\theta) \rangle = 0$ . Thus  $a \in S_I$ .

The proof chooses an arbitrary existential witness and does not require an isolated, unique, simple, or regular root. Images may overlap despite the parameter-cell tie rule, which does not alter set equality. A selected denominator may approach zero, but each nonzero value enters a finite weak threshold. Empty  $K_R$  makes both sides empty by Lemma A.1; empty or singleton intervals retain their actual points. For  $N = 1$ , beta is the empty tuple and (A24) is  $a_1 = -b/F_1$ .  $\square$

**Proposition A.8** (Root-event coefficient-volume certificate). *Under Assumptions 1 and 2, Propositions A.2 and A.6, Lemma A.5, Proposition A.7, and Lemma A.6, every interval  $I \subseteq \Theta$  satisfies*

$$\lambda_N(S_I) \leq (2R)^{N-1} \Gamma_{\text{piv}}(b, F; R) |I|. \quad (\text{A25})$$

*The conclusion retains all boundary, zero-denominator-limit, noninjective, tangent, multiple-root, infinite-fiber, empty-set, and  $N = 1$  regimes, with no chart-count, localization-count, multiplicity, or boundary factor.*

*Proof.* Fix  $j$  and set

$$A_{j,m} = \Psi_j(D_{j,m}), \quad A_j = \bigcup_{m=1}^{\infty} A_{j,m}.$$

Lemmas A.5 and A.6 make the  $A_{j,m}$  measurable and increasing. Continuity of measure from below and Proposition A.7 give

$$\begin{aligned} \lambda_N(A_j) &= \lim_{m \rightarrow \infty} \lambda_N(A_{j,m}) \\ &\leq (2R)^{N-1} \lim_{m \rightarrow \infty} \int_{I \cap E_{j,m}} V_j(\theta) d\theta. \end{aligned} \quad (\text{A26})$$

To avoid any undefined product  $0 \cdot (+\infty)$ , define

$$h_{j,m}(\theta) = \begin{cases} V_j(\theta), & \theta \in I \cap E_{j,m}, \\ 0, & \theta \notin I \cap E_{j,m}. \end{cases}$$

Proposition A.3 makes  $V_j$  finite on  $E_j$ , and Lemma A.4 makes the domains increase to  $E_j$ . Thus  $h_{j,m}$  are nonnegative Borel functions increasing to the function equal to  $V_j$  on  $I \cap E_j$  and zero elsewhere. Passing to the limit in their nonnegative integrals turns (A26) into

$$\lambda_N(A_j) \leq (2R)^{N-1} \int_{I \cap E_j} V_j(\theta) d\theta. \quad (\text{A27})$$

Every denominator boundary is recovered, and  $1/m$  leaves no tolerance or remainder.

By (A23), finite subadditivity, and (A27),

$$\lambda_N(S_I) \leq (2R)^{N-1} \sum_{j=1}^N \int_{I \cap E_j} V_j(\theta) d\theta. \quad (\text{A28})$$

On  $E_j$ , Propositions A.3 and A.5 give  $V_j = \min_i V_i \leq \Gamma_{\text{piv}}$ . The Borel cells are disjoint and cover  $K_R$ , so

$$\begin{aligned} \sum_{j=1}^N \int_{I \cap E_j} V_j(\theta) d\theta &\leq \Gamma_{\text{piv}}(b, F; R) \sum_{j=1}^N |I \cap E_j| \\ &= \Gamma_{\text{piv}}(b, F; R) |I \cap K_R| \\ &\leq \Gamma_{\text{piv}}(b, F; R) |I|. \end{aligned} \tag{A29}$$

Equations (A28)–(A29) prove (A25). Chart images may overlap, but the parameter cells do not; the sum costs one copy of  $|I \cap K_R|$ , not  $N|I|$ .

If  $K_R = \emptyset$ , Lemma A.1 gives  $S_I = \emptyset$  and  $\Gamma_{\text{piv}} = 0$ . Empty cells contribute zero. If  $I$  is empty or a singleton,  $|I| = 0$  and (A29) makes all chart integrals zero, so  $S_I$  is  $N$ -dimensionally null. For  $N = 1$  there is one cell and the beta factor is  $(2R)^0 = 1$ . The extended multiplicity already retains all remaining fiber pathologies.  $\square$

**Proposition A.9** (Measurable swept coefficient volume). *Under Assumptions 1 and 2, for every interval  $I \subseteq \Theta$ , the original-coordinate root set  $S_I$  in (A10) is analytic and Lebesgue measurable and satisfies (A25).*

*Proof.* Lemma A.5 verifies the descriptive-set interfaces and measurability. Proposition A.7 uses countable disjoint Lipschitz localization to obtain the extended multiplicity identity, determinant (A15), and beta-cube factor. Lemma A.6 proves both inclusions in (A23) in the original coefficient coordinates. Proposition A.8 then performs the increasing limit and uses the disjoint-cell identity (A29). The separate strict finiteness of  $\Gamma_{\text{piv}}$  remains the conclusion of Proposition A.2; the present result exports measurability and coefficient volume.  $\square$

#### A.4 Joint-Density Transfer and Ordered Suprema

**Proposition A.10** (Joint-density transfer for the swept root event). *Under Assumption 3 and Proposition A.9, every  $\mu \in \mathcal{D}_{N,R,\kappa}$  and every interval  $I \subseteq \Theta$  with  $|I| > 0$  satisfy*

$$\begin{aligned} \Pr_{\mu}(S_I) &\leq \kappa \lambda_N(S_I) \\ &\leq \kappa (2R)^{N-1} \Gamma_{\text{piv}}(b, F; R) |I| \\ &= \frac{A \Gamma_{\text{piv}}(b, F; R)}{2R} |I|. \end{aligned} \tag{A30}$$

Moreover,

$$\Pr_{\alpha \sim \mu} [\exists \theta \in I : \phi_{\alpha}(\theta) = 0] = \Pr_{\mu}(S_I). \tag{A31}$$

*These are ordinary probability statements under the completed law. They do not require independence, marginal densities, a conditioning event, a confidence parameter, or a small-interval threshold.*

*Proof.* Fix  $\mu$  and  $I$ , and write  $C = [-R, R]^N$  within this proof. Assumption 3 gives a nonnegative full joint density with

$$f_{\mu} = 0 \quad \lambda_N\text{-a.e. on } C^c, \quad f_{\mu} \leq \kappa \quad \lambda_N\text{-a.e. on } \mathbb{R}^N. \tag{A32}$$

Proposition A.9 makes  $S_I \subseteq C$  Lebesgue measurable. Absolute continuity and completion therefore give

$$\Pr_{\mu}(S_I) = \int_{S_I} f_{\mu}(\alpha) d\lambda_N(\alpha) \leq \kappa \lambda_N(S_I). \tag{A33}$$



This uses the cap on the full joint density directly; no coordinate is integrated out and arbitrary coordinate dependence contributes no term.

The volume certificate in Proposition A.9 applies to the identical set in the identical coefficient coordinates. Substituting it into (A33) gives the first two lines of (A30). Since  $R > 0$ ,

$$\kappa(2R)^{N-1} = \frac{\kappa(2R)^N}{2R} = \frac{A}{2R}, \quad (\text{A34})$$

which proves the last line without hiding, dropping, or absorbing a term.

Equation (A32) also gives  $\mu(C^c) = 0$ . The unrestricted root event in (A31) differs from  $S_I$  only by a subset of  $C^c$ ; that subset belongs to the completed law and has probability zero. This proves (A31) without changing the event on the law's support.

If  $K_R = \emptyset$ , Lemma A.1 makes  $S_I$  empty and the definition gives  $\Gamma_{\text{piv}} = 0$ . If  $S_I$  is empty for another reason, its volume and probability vanish. If  $\Gamma_{\text{piv}} = 0$  with nonempty  $K_R$ , the volume certificate forces  $\lambda_N(S_I) = 0$ , and (A33) forces zero probability. For  $N = 1$ ,  $(2R)^{N-1} = 1$  and  $A/(2R) = \kappa$ . Endpoints, cube faces, tangent and multiple roots, identically zero affine combinations, and finite or infinite fibers all remain in  $S_I$  and were handled before the density integration. The right side of (A30) may exceed one; it remains a valid bound and is not clipped.  $\square$

**Lemma A.7** (Nonemptiness threshold for the law class). *For  $N \geq 1$ ,  $R > 0$ ,  $\kappa \in (0, \infty)$ , and  $A = (2R)^N \kappa$ ,*

$$\mathcal{D}_{N,R,\kappa} \neq \emptyset \iff A \geq 1. \quad (\text{A35})$$

*The boundary  $A = 1$  is included, while  $A < 1$  makes the class empty.*

*Proof.* If  $\mu \in \mathcal{D}_{N,R,\kappa}$  and  $C = [-R, R]^N$ , then

$$\begin{aligned} 1 &= \int_{\mathbb{R}^N} f_\mu(\alpha) d\lambda_N(\alpha) \\ &= \int_C f_\mu(\alpha) d\lambda_N(\alpha) \\ &\leq \kappa \lambda_N(C) = \kappa(2R)^N = A. \end{aligned} \quad (\text{A36})$$

Thus nonemptiness implies  $A \geq 1$ , and correlation cannot alter this integral constraint.

Conversely, suppose  $A \geq 1$  and define

$$f_{\text{unif}}(\alpha) = (2R)^{-N} \mathbf{1}_{[-R,R]^N}(\alpha). \quad (\text{A37})$$

The cube has volume  $(2R)^N$ , so (A37) integrates to one and vanishes outside the cube. Moreover,

$$(2R)^{-N} \leq \kappa \iff 1 \leq \kappa(2R)^N = A. \quad (\text{A38})$$

Thus it defines an admissible law. At  $A = 1$ , (A38) is equality, proving the sharp boundary without adding  $A \geq 1$  as an assumption.  $\square$

**Lemma A.8** (Nonemptiness threshold for positive-length intervals). *For a compact interval  $\Theta \subset \mathbb{R}$ ,*

$$\{I \subseteq \Theta : I \text{ is an interval and } |I| > 0\} \neq \emptyset \iff |\Theta| > 0. \quad (\text{A39})$$

*Proof.* If the class on the left is nonempty, choose  $I$  in it. Monotonicity of Lebesgue measure gives  $0 < |I| \leq |\Theta|$ . Conversely, if  $|\Theta| > 0$ , the interval  $I = \Theta$  belongs to the class. Hence a zero-length compact interval has no positive-length subinterval, while a positive-length compact interval supplies one. Endpoint and openness conventions do not affect the equivalence.  $\square$

**Proposition A.11** (Ordered suprema with nonempty and vacuous semantics). *Under Assumptions 1, 2, and 3, and under Propositions A.2 and A.10, the following exhaustive alternatives hold. If  $A \geq 1$  and  $|\Theta| > 0$ , then*

$$0 \leq \sup_{\mu \in \mathcal{D}_{N,R,\kappa}} \sup_{\substack{I \subseteq \Theta, \text{ } I \text{ an interval} \\ |I| > 0}} \frac{\Pr_{\alpha \sim \mu}[\exists \theta \in I : \phi_\alpha(\theta) = 0]}{|I|} \leq \frac{A\Gamma_{\text{piv}}(b, F; R)}{2R} < \infty. \quad (\text{A40})$$

*The middle quantity is a finite nonnegative real. If  $A < 1$  or  $|\Theta| = 0$ , then, under  $\sup \emptyset = -\infty$ ,*

$$\sup_{\mu \in \mathcal{D}_{N,R,\kappa}} \sup_{\substack{I \subseteq \Theta, \text{ } I \text{ an interval} \\ |I| > 0}} \frac{\Pr_{\alpha \sim \mu}[\exists \theta \in I : \phi_\alpha(\theta) = 0]}{|I|} = -\infty \leq \frac{A\Gamma_{\text{piv}}(b, F; R)}{2R}. \quad (\text{A41})$$

*The value in (A41) is a vacuous extended-real consequence, not a finite nonnegative capacity. Proposition A.10 remains a universal pairwise statement in every branch.*

*Proof.* Suppose first that  $A \geq 1$  and  $|\Theta| > 0$ . Lemmas A.7 and A.8 show that both index classes are nonempty. For fixed  $\mu$ , Proposition A.10 gives for each indexed interval

$$0 \leq \frac{\Pr_{\alpha \sim \mu}[\exists \theta \in I : \phi_\alpha(\theta) = 0]}{|I|} \leq \frac{A\Gamma_{\text{piv}}(b, F; R)}{2R}, \quad (\text{A42})$$

where division is legal because  $|I| > 0$ . First take the supremum over intervals for this fixed law:

$$0 \leq \sup_{\substack{I \subseteq \Theta, \text{ } I \text{ an interval} \\ |I| > 0}} \frac{\Pr_{\alpha \sim \mu}[\exists \theta \in I : \phi_\alpha(\theta) = 0]}{|I|} \leq \frac{A\Gamma_{\text{piv}}(b, F; R)}{2R}. \quad (\text{A43})$$

Then take the supremum over admissible laws. This preserves the stated inner-then-outer order and is not a union bound over either index. Proposition A.2 gives  $0 \leq \Gamma_{\text{piv}} < \infty$  in both feasible-set branches. Since  $A$  is finite and positive and  $R > 0$ , the right side is finite. Nonempty index classes and nonnegative ratios make the ordered supremum nonnegative. If  $\Gamma_{\text{piv}} = 0$ , every ratio is zero and the supremum is zero.

If  $A < 1$ , Lemma A.7 makes the outer class empty, so its supremum is  $-\infty$ , irrespective of the inner class. If  $|\Theta| = 0$ , Lemma A.8 makes every inner class empty. When laws exist, the outer supremum of the constant value  $-\infty$  remains  $-\infty$ ; when the law class is also empty, the outer empty supremum has the same value. This covers both single-empty and double-empty branches. At  $A = 1$ , the uniform cube law from (A37) exists, so the boundary belongs to (A40) whenever  $|\Theta| > 0$ . No branch condition is inserted into the pairwise theorem.  $\square$

**Proposition A.12** (Density conversion and branch-qualified uniform consequence). *Under Assumptions 1, 2, and 3, every admissible law and positive-length interval satisfy (A30)–(A31), while the ordered law-then-interval expression has the semantics in (A40)–(A41), with the interval supremum taken first. A finite nonnegative capacity interpretation is valid only when  $A \geq 1$  and  $|\Theta| > 0$ .*

*Proof.* Proposition A.9 supplies the measurable original-coordinate event and its coefficient-volume bound. Proposition A.10 integrates that same event against the same capped full joint density and proves the pairwise result with the algebraic identity (A34). The sharp index classifications are Lemmas A.7 and A.8. Proposition A.11 then takes the two suprema in their declared order, using strict finiteness only from Proposition A.2. It retains  $-\infty$  whenever an index class is empty and does not reinterpret that value as a capacity.  $\square$

## A.5 Exact Scale-Stress Conditioning

**Lemma A.9** (Root-feasible set for the scale-stress family). *For  $0 < \delta \leq 1$ , let*

$$\Theta = [-1, 1], \quad b_\delta(\theta) = 0, \quad F_\delta(\theta) = \left(1, \frac{\theta}{\delta}\right), \quad R = 1.$$

*This family satisfies Assumptions 1 and 2, its feasible set is  $K_1 = \Theta$ , and its first feature is a legal pivot everywhere on  $K_1$ .*

*Proof.* For fixed  $\delta > 0$ , the three scalar functions are  $C^1$  polynomials in  $\theta$  with a vacuous chain, and the first feature is identically one. Thus the two stated assumptions hold. The definition of the feasible set gives

$$\begin{aligned} K_1 &= \{\theta \in \Theta : |b_\delta(\theta)| \leq \|F_\delta(\theta)\|_1\} \\ &= \left\{ \theta \in \Theta : 0 \leq 1 + \frac{|\theta|}{\delta} \right\} = \Theta. \end{aligned} \tag{A44}$$

Both inclusions are literal: the inequality holds because  $\delta > 0$ , and the set comprehension already restricts to  $\Theta$ . Hence  $K_1$  contains 0 and both endpoints. Since  $F_{\delta,1}(\theta) = 1$ , coordinate one is nonzero everywhere. This realizes the legal-pivot conclusion of Proposition A.2 without imposing a global-pivot condition on general families.  $\square$

**Lemma A.10** (Extended pivot speeds for the scale-stress family). *Under Lemma A.9, for every  $0 < \delta \leq 1$ ,*

$$V_1(\theta) = \frac{1}{\delta} \quad (\theta \in \Theta), \quad V_2(\theta) = \begin{cases} \frac{\delta}{\theta^2}, & \theta \neq 0, \\ +\infty, & \theta = 0. \end{cases} \tag{A45}$$

*Proof.* Write  $F_{\delta,1}(\theta) = 1$  and  $F_{\delta,2}(\theta) = \theta/\delta$ . The first coordinate is nonzero on all of  $\Theta$ . With  $N = 2$  and  $R = 1$ ,

$$V_1(\theta) = \left| \left( \frac{b_\delta}{F_{\delta,1}} \right)'(\theta) \right| + \left| \left( \frac{F_{\delta,2}}{F_{\delta,1}} \right)'(\theta) \right|.$$

Keeping every quotient-rule numerator and denominator visible,

$$\left( \frac{b_\delta}{F_{\delta,1}} \right)' = \frac{0 \cdot 1 - 0 \cdot 0}{1^2} = 0, \quad \left( \frac{F_{\delta,2}}{F_{\delta,1}} \right)' = \frac{(1/\delta) \cdot 1 - (\theta/\delta) \cdot 0}{1^2} = \frac{1}{\delta}.$$

Since  $\delta > 0$ , this proves  $V_1 = 1/\delta$  everywhere, including zero and both endpoints.

If  $\theta \neq 0$ , then  $F_{\delta,2}(\theta) \neq 0$  and

$$V_2(\theta) = \left| \left( \frac{b_\delta}{F_{\delta,2}} \right)'(\theta) \right| + \left| \left( \frac{F_{\delta,1}}{F_{\delta,2}} \right)'(\theta) \right|.$$

The two quotient derivatives are

$$\left( \frac{b_\delta}{F_{\delta,2}} \right)' = \frac{0 \cdot (\theta/\delta) - 0 \cdot (1/\delta)}{(\theta/\delta)^2} = 0$$

and

$$\begin{aligned} \left( \frac{F_{\delta,1}}{F_{\delta,2}} \right)' &= \frac{0 \cdot (\theta/\delta) - 1 \cdot (1/\delta)}{(\theta/\delta)^2} \\ &= -\frac{1/\delta}{\theta^2/\delta^2} = -\frac{\delta}{\theta^2}. \end{aligned}$$

Thus  $V_2(\theta) = \delta/\theta^2$  off zero. At zero, the second feature vanishes, so its quotient is not differentiated; the definition instead sets  $V_2(0) = +\infty$ . Endpoint derivatives use the same  $C^1$  convention.  $\square$

**Proposition A.13** (Exact scale-stress pivot conditioning). *Under Proposition A.2 and Lemmas A.9 and A.10, the following holds for every  $0 < \delta \leq 1$ , including  $\delta = 1$ :*

$$\Gamma_{\text{piv}}(b_\delta, F_\delta; 1) = \frac{1}{\delta}. \quad (\text{A46})$$

*Proof.* For every  $\theta \in \Theta$ ,

$$\min\{V_1(\theta), V_2(\theta)\} \leq V_1(\theta) = \frac{1}{\delta}. \quad (\text{A47})$$

At  $\theta = 0$ ,

$$\min\{V_1(0), V_2(0)\} = \min\left\{\frac{1}{\delta}, +\infty\right\} = \frac{1}{\delta}. \quad (\text{A48})$$

Lemma A.9 gives  $K_1 = \Theta$  and  $0 \in K_1$ . Consequently,

$$\frac{1}{\delta} = \min\{V_1(0), V_2(0)\} \leq \sup_{\theta \in K_1} \min\{V_1(\theta), V_2(\theta)\} \leq \frac{1}{\delta}. \quad (\text{A49})$$

The nonempty branch of the definition of  $\Gamma_{\text{piv}}$  turns (A49) into (A46).

At zero only pivot one has finite speed. Away from zero the two speeds tie exactly when

$$\frac{1}{\delta} = \frac{\delta}{\theta^2}, \quad \text{that is, } \theta = \pm\delta.$$

These points lie in  $[-1, 1]$  for the full range  $0 < \delta \leq 1$ ; for  $\delta = 1$  they are the endpoints. The least-index convention chooses pivot one at a tie but does not change the pointwise minimum or supremum. The coefficient in (A46) is exactly one. This is a deterministic conditioning identity and asserts no additional probability bound.  $\square$

## A.6 Exact Affine-Monic Recovery

**Proposition A.14** (Original-space monic presentation). *Fix  $d \geq 1$ ,  $R > 0$ ,  $\kappa > 0$ , and a compact interval  $\Theta \subset \mathbb{R}$ . Under Assumption 3, let  $\mu$  be any Borel probability law on  $\mathbb{R}^d$  with full joint density supported almost everywhere on  $[-R, R]^d$  and bounded almost everywhere by  $\kappa$ . Define*

$$p_\alpha(\theta) = \theta^d + \sum_{k=0}^{d-1} \alpha_k \theta^k, \quad b_d(\theta) = \theta^d, \quad F_d(\theta) = (1, \theta, \dots, \theta^{d-1}). \quad (\text{A50})$$

Then for every  $\alpha \in \mathbb{R}^d$  and  $\theta \in \Theta$ ,

$$b_d(\theta) + \langle \alpha, F_d(\theta) \rangle = p_\alpha(\theta) \quad (\text{A51})$$

in the original  $d$ -dimensional lower-coefficient space. This family satisfies Assumptions 1 and 2, and its descriptor tuple is

$$q = 0, \quad M = 0, \quad \Delta_{\text{rnd}} = d - 1, \quad \Delta_{\text{aff}} = d, \quad N = d, \quad A = (2R)^d \kappa. \quad (\text{A52})$$

*Proof.* Index the features from one to  $d$ , so  $(F_d)_i(\theta) = \theta^{i-1}$ . Then

$$\langle \alpha, F_d(\theta) \rangle = \sum_{i=1}^d \alpha_{i-1} \theta^{i-1} = \sum_{k=0}^{d-1} \alpha_k \theta^k,$$

and adding  $b_d(\theta) = \theta^d$  proves (A51). This is equality for the same  $\alpha = (\alpha_0, \dots, \alpha_{d-1})$ . The leading coefficient one remains deterministic in  $b_d$ ; there is no additional coordinate, transformed law, singular augmentation, or coefficient-space change.

For Assumption 1, take the vacuous-chain presentation  $q = 0$ , with no  $\eta$  and no chain polynomials. The convention gives  $M = 0$ , and the output polynomials are

$$Q_0(\theta) = \theta^d, \quad Q_i(\theta) = \theta^{i-1} \quad (1 \leq i \leq d).$$

They are  $C^1$  on every compact interval and, in the same scalar convention  $p = 1$ ,

$$\max_{1 \leq i \leq d} \deg Q_i = d - 1, \quad \max_{0 \leq i \leq d} \deg Q_i = d.$$

This includes  $d = 1$ , for which the only random feature is the degree-zero constant. Hence  $\Delta_{\text{rnd}} = d - 1$ ,  $\Delta_{\text{aff}} = d$ , and there are  $N = d$  random lower coefficients.

For Assumption 2, the first feature obeys  $(F_d)_1(\theta) = 1$  everywhere. Hence  $F_d(\theta) \neq 0$ , including at zero and at both endpoints. The law hypothesis is the full joint-density cap in the same coefficient space, with only almost-everywhere cube support. Substituting  $N = d$  into  $A = (2R)^N \kappa$  completes (A52). No coordinate independence has been used or inferred.  $\square$

**Lemma A.11** (Constant-pivot velocity on the closed unit region). *Under Proposition A.14, the constant feature is a legal pivot at every  $\theta$ . If  $d = 1$ , its speed is exactly*

$$V_{\text{const}}(\theta) = 1. \tag{A53}$$

If  $d \geq 2$  and  $|\theta| \leq 1$ , then

$$V_{\text{const}}(\theta) = d|\theta|^{d-1} + R \sum_{k=1}^{d-1} k|\theta|^{k-1} \leq d + R \frac{d(d-1)}{2}. \tag{A54}$$

Thus the displayed upper bound holds for every  $d \geq 1$  throughout the closed unit region, with exact speed one for the entire  $d = 1$  branch.

*Proof.* The authoritative pivot definition gives

$$V_{\text{const}}(\theta) = \left| \left( \frac{b_d}{1} \right)'(\theta) \right| + R \sum_{k=1}^{d-1} \left| \left( \frac{\theta^k}{1} \right)'(\theta) \right|. \tag{A55}$$

There are exactly  $d - 1$  nonpivot features; the deterministic leading monomial appears only in the first term. For  $d = 1$ , (A55) is

$$|\theta'| + R \sum_{k=1}^0 (\dots) = 1,$$

which proves (A53).

For  $d \geq 2$ , direct differentiation gives

$$|(d\theta^{d-1})| = d|\theta|^{d-1}, \quad |(k\theta^{k-1})| = k|\theta|^{k-1},$$

for both signs of  $\theta$ , proving equality in (A54). At  $\theta = 0$  this is read directly: the leading derivative is zero, the  $k = 1$  feature derivative is one, and all  $k \geq 2$  derivatives are zero. No  $0^0$  convention is used.

For  $|\theta| \leq 1$ , every displayed power is at most one, and so

$$V_{\text{const}}(\theta) \leq d + R \sum_{k=1}^{d-1} k.$$

The arithmetic remains explicit:

$$2 \sum_{k=1}^{d-1} k = \sum_{k=1}^{d-1} (k + (d - k)) = d(d - 1), \quad \sum_{k=1}^{d-1} k = \frac{d(d - 1)}{2}. \quad (\text{A56})$$

This proves (A54). Both points  $\theta = 1$  and  $\theta = -1$  stay in this constant chart; they are not reassigned to the top chart.  $\square$

**Proposition A.15** (Top-pivot velocity and global monic conditioning). *Under Proposition A.2, Proposition A.14, and Lemma A.11, if  $d \geq 2$  and  $|\theta| > 1$ , then the top feature  $(F_d)_d(\theta) = \theta^{d-1}$  is nonzero and*

$$V_{\text{top}}(\theta) = 1 + R \sum_{m=1}^{d-1} \frac{m}{|\theta|^{m+1}} \leq 1 + R \frac{d(d-1)}{2} \leq d + R \frac{d(d-1)}{2}. \quad (\text{A57})$$

For every  $d \geq 1$ ,  $R > 0$ , and compact interval  $\Theta$ , the exact feasible set of the restricted monic family is

$$K_R = \left\{ \theta \in \Theta : |\theta|^d \leq R \sum_{k=0}^{d-1} |\theta|^k \right\}, \quad (\text{A58})$$

and

$$\Gamma_{\text{piv}}(b_d, F_d; R) \leq d + R \frac{d(d-1)}{2}. \quad (\text{A59})$$

The bound is independent of  $\Theta$ .

*Proof.* Assume first  $d \geq 2$  and  $|\theta| > 1$ . Then  $\theta \neq 0$ , so the top feature is a legal pivot. Its offset ratio is

$$\frac{b_d(\theta)}{(F_d)_d(\theta)} = \frac{\theta^d}{\theta^{d-1}} = \theta, \quad \left| \left( \frac{b_d}{(F_d)_d} \right)'(\theta) \right| = 1. \quad (\text{A60})$$

The lower nonpivot exponents are  $k = 0, \dots, d-2$ . Reindex them by

$$m = d - 1 - k \in \{1, \dots, d-1\}, \quad k - (d-1) = -m. \quad (\text{A61})$$

Then

$$\frac{\theta^k}{\theta^{d-1}} = \theta^{-m}, \quad |(\theta^{-m})'| = |-m\theta^{-m-1}| = \frac{m}{|\theta|^{m+1}}. \quad (\text{A62})$$

Substituting (A60)–(A62) into the definition of  $V_j$  gives equality in (A57) for both signs of  $\theta$ . Since  $|\theta| > 1$ , each  $m/|\theta|^{m+1} \leq m$ ; equation (A56) gives the first inequality, and  $1 \leq d$  gives the second. There is no remainder or large-radius threshold.

Substitution of the monic objects into the definition of  $K_R$  gives

$$|b_d(\theta)| = |\theta|^d, \quad \|F_d(\theta)\|_1 = \sum_{k=0}^{d-1} |\theta|^k = \sum_{k=0}^{d-1} |\theta|^k,$$

which is (A58). At every point of this set take the minimum over legal pivots. When  $d = 1$ , the constant feature is the sole feature and Lemma A.11 gives

$$\min_{j: (F_1)_j(\theta) \neq 0} V_j(\theta) = 1 = d + R \frac{d(d-1)}{2}. \quad (\text{A63})$$

When  $d \geq 2$  and  $|\theta| \leq 1$ , the constant feature is legal and

$$\min_{j: (F_d)_j(\theta) \neq 0} V_j(\theta) \leq V_{\text{const}}(\theta) \leq d + R \frac{d(d-1)}{2}. \quad (\text{A64})$$

When  $d \geq 2$  and  $|\theta| > 1$ , the top feature is legal and

$$\min_{j:(F_d)_j(\theta) \neq 0} V_j(\theta) \leq V_{\text{top}}(\theta) \leq d + R \frac{d(d-1)}{2}. \quad (\text{A65})$$

At  $|\theta| = 1$ , only (A64) is used, so there is no duplicated boundary or chart-count cost.

If  $K_R$  is nonempty, taking its defining supremum in (A63)–(A65) proves (A59). If it is empty, the definition and Proposition A.2 give  $\Gamma_{\text{piv}} = 0$ . A chart region may itself be empty and then contributes nothing.

For  $d = 2$ , the two branches reduce explicitly to

$$\begin{aligned} V_{\text{const}}(\theta) &= 2|\theta| + R \leq 2 + R \quad (|\theta| \leq 1), \\ V_{\text{top}}(\theta) &= 1 + \frac{R}{|\theta|^2} \leq 1 + R \leq 2 + R \quad (|\theta| > 1). \end{aligned}$$

Thus the first nontrivial dimension is covered. The right side of (A59) depends only on  $d$  and  $R$ , not the location, diameter, or endpoints of  $\Theta$ .  $\square$

**Proposition A.16** (Positive-length monic probability transfer). *Under Assumption 3, Propositions A.14, A.15, and A.10, let  $I \subset \mathbb{R}$  be a bounded interval with  $|I| > 0$ . For the same arbitrary full joint law  $\mu$  on the original  $d$  lower coefficients,*

$$\Pr_{\alpha \sim \mu} [\exists \theta \in I : p_\alpha(\theta) = 0] \leq \kappa(2R)^{d-1} \left( d + R \frac{d(d-1)}{2} \right) |I|. \quad (\text{A66})$$

Equivalently,

$$\kappa(2R)^{d-1} \left( d + R \frac{d(d-1)}{2} \right) |I| = \frac{A}{2R} \left( d + R \frac{d(d-1)}{2} \right) |I|. \quad (\text{A67})$$

*Proof.* Set  $a = \inf I$  and  $c = \sup I$ . Boundedness makes both finite. If  $a = c$ , the between-points property of an interval would make  $I$  contain at most one point and hence have length zero, contrary to the hypothesis. Thus  $a < c$ , and the interval property gives  $\bar{I} = [a, c]$ . Use the nondegenerate compact localization  $\Theta = [a, c]$ .

Every hypothesis of Proposition A.10 now holds for one fixed admissible pair. Proposition A.14 gives  $N = d \geq 1$ , the  $C^1$  vacuous-chain presentation on this  $\Theta$ , and no-forced-root nondegeneracy from the constant feature. The given law has the capped full joint density in the same original coefficient space, with almost-everywhere support on  $[-R, R]^d$  and no independence premise. The interval  $I$  lies in  $\Theta$  and has positive length. Proposition A.15 applies to this restriction and gives

$$\Gamma_{\text{piv}}(b_d, F_d; R) \leq d + R \frac{d(d-1)}{2},$$

independently of the chosen localization. Finally, (A51) identifies the affine event with the polynomial event for the same  $\alpha, \theta, I$ , and  $\mu$ .

Applying (A30) with  $N = d$  therefore gives

$$\begin{aligned} \Pr_{\alpha \sim \mu} [\exists \theta \in I : p_\alpha(\theta) = 0] &= \Pr_{\alpha \sim \mu} [\exists \theta \in I : b_d(\theta) + \langle \alpha, F_d(\theta) \rangle = 0] \\ &\leq \kappa(2R)^{d-1} \Gamma_{\text{piv}}(b_d, F_d; R) |I| \\ &\leq \kappa(2R)^{d-1} \left( d + R \frac{d(d-1)}{2} \right) |I|, \end{aligned}$$

which is (A66). Because  $R > 0$ ,

$$\kappa(2R)^{d-1} = \frac{\kappa(2R)^d}{2R} = \frac{A}{2R},$$

proving (A67) while preserving the requested  $\kappa$  form.

If  $I \cap K_R = \emptyset$ , Lemma A.1 rules out roots on  $I$  for every coefficient in the cube. Almost-everywhere cube support assigns probability one to that cube, so the unrestricted event has probability zero. This includes empty  $K_R$  and shows that no nonemptiness condition entered the argument. The right side of (A66) may exceed one; no clipping, small-interval condition, or probability-mode change is made. The ordered supremum and its capacity interpretation are not used in this specialization.  $\square$

**Proposition A.17** (Complete exact monic baseline). *Under Assumption 3, Proposition A.14, and Proposition A.16, every bounded interval  $I \subset \mathbb{R}$ , including  $|I| = 0$ , satisfies*

$$\Pr_{\alpha \sim \mu} [\exists \theta \in I : p_\alpha(\theta) = 0] \leq \kappa(2R)^{d-1} \left( d + R \frac{d(d-1)}{2} \right) |I|. \quad (\text{A68})$$

*For a singleton interval, its coefficient root set is a proper affine hyperplane of  $\mathbb{R}^d$  and has both  $d$ -dimensional Lebesgue measure and  $\mu$ -probability zero, including  $d = 1$ .*

*Proof.* Positive-length intervals are covered by Proposition A.16. It remains to treat zero length. If a real interval contained two distinct points  $x < y$ , it would contain  $(x, y)$  and have length at least  $y - x > 0$ . Hence a zero-length interval is empty or a singleton. The root event is empty when  $I = \emptyset$ .

Suppose  $I = \{\theta_0\}$ . Equation (A51) gives the coefficient root set

$$\begin{aligned} H_{\theta_0} &= \{\alpha \in \mathbb{R}^d : p_\alpha(\theta_0) = 0\} \\ &= \left\{ \alpha \in \mathbb{R}^d : \alpha_0 = -\theta_0^d - \sum_{k=1}^{d-1} \alpha_k \theta_0^k \right\}. \end{aligned} \quad (\text{A69})$$

The coefficient of  $\alpha_0$  is one. Equivalently, this is an affine level set with nonzero normal  $F_d(\theta_0) = (1, \theta_0, \dots, \theta_0^{d-1})$ . It is therefore proper for every  $\theta_0$ , including zero and  $d = 1$ .

We prove nullity in these coordinates. If  $d = 1$ , then  $H_{\theta_0} = \{-\theta_0\}$ . For every  $\varepsilon > 0$ , this point is covered by an interval of length  $\varepsilon$ , so its outer measure is at most  $\varepsilon$ ; letting  $\varepsilon \downarrow 0$  gives  $\lambda_1(H_{\theta_0}) = 0$ .

If  $d \geq 2$ , write  $\beta = (\alpha_1, \dots, \alpha_{d-1}) \in \mathbb{R}^{d-1}$ . The set in (A69) is the graph of the continuous affine function

$$\beta \mapsto -\theta_0^d - \sum_{k=1}^{d-1} \beta_k \theta_0^k.$$

It is closed and Lebesgue measurable. For fixed  $\beta$ , its  $\alpha_0$ -section is a singleton and has one-dimensional measure zero. Nonnegative product integration of its indicator yields

$$\begin{aligned} \lambda_d(H_{\theta_0}) &= \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}} \mathbf{1}_{H_{\theta_0}}(\alpha_0, \beta) d\alpha_0 d\beta \\ &= \int_{\mathbb{R}^{d-1}} 0 d\beta = 0. \end{aligned} \quad (\text{A70})$$

Thus nullity is derived rather than assumed.



The full joint density now gives

$$\Pr_{\alpha \sim \mu} [p_\alpha(\theta_0) = 0] = \mu(H_{\theta_0}) = \int_{H_{\theta_0}} f_\mu d\lambda_d = 0. \quad (\text{A71})$$

No marginal density or independence is used. Almost-everywhere cube support creates no exceptional mass because the law is absolutely continuous. The right side of (A68) is also zero when  $|I| = 0$ , completing the empty and singleton branches.  $\square$

**Proposition A.18** (Exact affine-monic specialization and baseline bridge). *Under Assumption 3, Proposition A.2, and the pairwise probability result in Proposition A.10, the monic family in (A50) has the original-space identity and descriptors (A51)–(A52), obeys the conditioning bound (A59) for every compact restriction, and satisfies the all-bounded- interval probability bound (A68). The coefficient is literally*

$$\kappa(2R)^{d-1} \left( d + R \frac{d(d-1)}{2} \right),$$

*with no hidden constant, transformed probability, random leading coefficient, independence condition, or conservative remainder.*

*Proof.* Proposition A.14 keeps the leading coefficient in  $b_d$ , keeps exactly  $d$  lower coefficients in the random vector, proves  $b_d + \langle \alpha, F_d \rangle = p_\alpha$ , checks both deterministic assumptions, and computes the descriptor tuple. Lemma A.11 computes the closed-unit-region speed, including  $d = 1$ , zero, both signs, and  $|\theta| = 1$ . Proposition A.15 computes the top-pivot speed with the reindexing (A61), takes the pointwise minimum over legal pivots, and then takes the supremum over the exact  $K_R$ , proving (A59) with no localization dependence.

For positive-length bounded intervals, Proposition A.16 checks the compact localization and every premise of the pairwise probability result, then uses the same objects and same law to obtain (A66) and the exact algebra (A67). Proposition A.17 adds empty and singleton intervals by the explicit nullity calculation (A69)–(A71). No ordered-supremum or capacity branch is used. The bridge has ordinary probability for each arbitrary full joint law, static uniformity over bounded intervals, the scalar pivot-variation and full  $d$ -dimensional Lebesgue modes, and no auxiliary parameter or hidden constant.

This fixed-family bridge does not give polynomial general-instance control of  $\Gamma_{\text{piv}}$  in the descriptors

$$q, \quad M, \quad \Delta_{\text{rnd}}, \quad \Delta_{\text{aff}},$$

or in any other general Pfaffian presentation data. That source-direction gap remains unresolved.  $\square$

## A.7 Proof of the Main Theorem

*Proof of Theorem 3.1.* Proposition A.2 proves, from Assumptions 1 and 2, that every cube-supported root is feasible. It also proves, including the empty- $K_R$  branch, that

$$0 \leq \Gamma_{\text{piv}}(b, F; R) < \infty.$$

This is the first assertion of Theorem 3.1.

Proposition A.6 constructs the disjoint Borel pivot cells, their complete finite-denominator exhaustion, and the original-coordinate graph maps from that finite-conditioning interface. Using

precisely those charts, Proposition A.9 proves measurability of the root event and its coefficient-volume bound. Proposition A.12 combines that volume result with the primitive full joint-density cap, while using Proposition A.2 directly for strict finiteness. Its pairwise statement is

$$\Pr_{\alpha \sim \mu} [\exists \theta \in I : \phi_\alpha(\theta) = 0] \leq \kappa(2R)^{N-1} \Gamma_{\text{piv}}(b, F; R) |I| = \frac{A \Gamma_{\text{piv}}(b, F; R)}{2R} |I|,$$

which is (T1). Its sharp law- and interval-index classifications then give (T2) with a finite nonnegative value exactly when  $A \geq 1$  and  $|\Theta| > 0$ , and with the literal value  $-\infty$  in every empty-index branch.

Proposition A.13, based on the common finite-conditioning result and the two explicit specialized pivot speeds, proves (T3) for the entire range  $0 < \delta \leq 1$ . Proposition A.18 uses the same common root-feasibility result and only the pairwise probability statement of Proposition A.10. It proves the two-pivot bound (T4), preserves the deterministic leading coefficient, and derives (T5) for positive-length intervals before completing empty and singleton intervals by full-joint-density nullity.

All coefficients in these cited propositions are displayed literally: there is no hidden constant, probability-mode conversion, horizon upgrade, independence premise, or suppressed norm change. Their fixed-family scope also leaves the stated absence of a polynomial general Pfaffian-format bound intact. This proves every clause of Theorem 3.1.  $\square$